Inherited Ecosystem Structure: A Spectral-Geometric Framework for Reference-Free Genomic Relatedness Detection

yalumba project

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ABSTRACT

We propose that biological relatedness between individuals can be detected by comparing the *organizational structure* of their genomes rather than their *sequence content*. In our framework, each genome is modeled as a multi-scale ecosystem of interacting motif modules—groups of co-occurring k-mers that form coherent functional or structural units within sequencing reads. We analyze this ecosystem via spectral geometry, constructing a labeled interaction graph whose nodes are modules and whose edges encode co-occurrence strength, then characterizing the graph through its Laplacian spectrum, heat kernel signatures, and ecological role assignments. Relatedness between two individuals is detected through *structural correspondence*: a continuous coherence field $\Phi : G_A \to G_B$ that maps the ecosystem of one genome into the ecosystem of another, scored by how well this mapping preserves local neighbourhood geometry. This is fundamentally different from identity-by-descent (IBD) methods, which compare shared sequence tokens—identical k-mers, shared haplotype segments, or concordant variant calls. We have built a complete from-scratch compute stack (12 packages, ∼15,000 lines of TypeScript, custom GPU kernel pipeline) and tested more than 50 algorithms across three datasets: a synthetic family trio, the GIAB Ashkenazi trio, and the CEPH 1463 Platinum Pedigree (a three-generation, six-member family that constitutes the hardest benchmark for reference-free methods). Three key findings emerge from this systematic empirical program. **First**, token-matching methods—from naive k-mer Jaccard through sophisticated rare-run scanning—achieve strong separation on related versus unrelated pairs but converge toward classical IBD proxies: the best token method, rare-run P90, achieves +583% separation on CEPH 1463 and is mathematically equivalent to reference-free IBD without alignment. **Second**, structural methods—module persistence (+3.36%), coalition transfer (+1.97%), spectral ecology (+4.93%), and module stability (+4.90% average but failing the weakest pair)—detect a genuinely different signal, inherited organizational patterns, but this signal is 100× weaker than the rare-run signal and is confounded by sequencing coverage variation. **Third**, the gap between structural and token methods has narrowed dramatically over six algorithm generations in the spectral ecology family: from complete failure (−26.87% weakest gap in v1) through the spouse-confound breakthrough (v4, spouses dropped from rank 1 to rank 8) to near-parity (−0.61% weakest gap in v7 via Sinkhorn-regularized transport with cooperative stability weighting). We identify three theoretical barriers to further progress—coverage confounding, transport symmetry, and outlier sensitivity—and propose solutions rooted in optimal transport (Gromov–Wasserstein distance between ecosystem metric spaces), algebraic topology (persistent homology of co-occurrence filtrations), and differential geometry (gauge-theoretic field comparison via connection curvature). These three directions represent qualitative mathematical leaps rather than incremental parameter tuning, and we argue that the strongest case for the ecosystem approach will come not from beating rare-run P90 at parent–child detection but from detecting relationships or organizational structures that token-based methods fundamentally cannot access. This paper serves as a theoretical position statement for external feedback: we present the complete intellectual framework, the empirical trajectory, the open problems, and the proposed path forward.

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1 Introduction

1.1 The Reference-Free Relatedness Problem

Determining whether two individuals share recent common ancestors is a foundational problem in human genetics. The standard pipeline requires multiple computationally expensive steps:

1. **Alignment** to a reference genome (BWA-MEM, minimap2): hours per sample, reference-dependent.
2. **Variant calling** (GATK HaplotypeCaller, DeepVariant): hours per sample, model-dependent.
3. **Phasing** (SHAPEIT, Eagle): population-level, requires reference panels.
4. **IBD detection** (GERMLINE [10], Refined IBD [3], IBIS [4]): segment matching on phased haplotypes.

Each step introduces its own error modes and computational cost. Alignment is reference-biased [14]. Variant calling has systematic blind spots in repetitive regions. Phasing requires large reference panels and introduces switch errors. The result is a pipeline that works well for well-characterized populations with good reference genomes, but degrades for under-represented populations, non-model organisms, and metagenomic samples where no reference exists at all.

Can we detect relatedness directly from raw sequencing reads, without alignment, variant calling, or phasing?

This is the reference-free relatedness problem. Prior work has shown that k-mer sketching can estimate genomic distance (Mash [19]), that alignment-free phylogenetics is feasible [2], and that reference-free relatedness estimation is possible in principle [9]. Our contribution is a theoretical framework and systematic empirical program that goes beyond k-mer counting to propose a fundamentally different basis for comparison: the *organizational structure* of the genome as a multi-scale ecosystem.

1.2 Two Paradigms for Genomic Comparison

We distinguish two paradigms that differ not in implementation details but in what *mathematical object* is compared between individuals.

1.2.1 The Token Paradigm

Genomes are bags of discrete tokens: k-mers, variants, haplotype segments. Relatedness is measured by shared tokens. This is the classical view underlying all IBD-based methods.

- **Definition**

Token comparison. Given two genomes G_A and G_B represented as sets of tokens T_A and T_B, relatedness is a function $f(T_A \cap T_B, T_A \cup T_B)$. The Jaccard coefficient, containment index, MinHash estimate, and IBD segment count are all instances of this pattern.

Token methods are intuitive, well-understood, and effective. They have a clear failure mode: they require tokens to be *identical* across individuals. In genomic data, heterozygosity means that k-mers within a shared IBD segment may differ at SNP positions. At $k = 21$, a single heterozygous SNP destroys 21 k-mers. With a SNP rate of $\sim 0.1\%$ (one per 1,000 bases), the probability that a 21-mer is identical between two haplotypes is approximately $(1-0.001)^{21} \approx 0.979$. This means $\sim$2% of k-mers are disrupted by heterozygosity even within truly shared segments.

1.2.2 The Ecosystem Paradigm (Our Proposal)

Genomes are organized systems of interacting elements. Relatedness is measured by shared *organizational structure*. This does not require any element to be identical across individuals—only that the *pattern of organization* is similar.

- **Definition**

Ecosystem comparison. Given two genomes G_A and G_B represented as labeled interaction graphs $\mathcal{G}_A = (V_A, E_A, w_A, f_A)$ and $\mathcal{G}_B = (V_B, E_B, w_B, f_B)$, relatedness is a function of the structural correspondence between $\mathcal{G}_A$ and $\mathcal{G}_B$: how well can the local neighbourhood geometry of one be mapped to the other?

The ecosystem paradigm is more robust to noise (a few disrupted k-mers do not destroy the organizational structure) but harder to measure (structural correspondence is a richer and more ambiguous quantity than set intersection). The central question of this research program is whether the additional robustness translates into practical advantage.

1.3 Symbiogenesis as Computational Metaphor

Our framework draws on Lynn Margulis's theory of symbiogenesis [15]: the idea that complex life arose not through gradual mutation and selection alone, but through the *merging*

of simpler organisms into cooperative wholes. Mitochondria were once free-living bacteria; chloroplasts were once cyanobacteria. The eukaryotic cell is a consortium, not a monolith.

We apply this metaphor to genomic inheritance. Each haplotype block inherited from a parent is an "organism" carrying its internal structure—its pattern of motif modules, their co-occurrence relationships, their spectral geometry. During meiosis, these organisms recombine: maternal and paternal haplotype blocks interleave to form a new consortium. The child's genome is a *symbiogenetic merger* of parental organizational units.

> ▶ **Key Finding**
>
> **The symbiogenesis metaphor predicts a specific testable quantity**: related individuals should share not just individual genomic elements but the *cooperative structure* among those elements. Two genomes that inherited the same haplotype block from a common ancestor should exhibit the same local arrangement of motif modules—the same co-occurrence edges, the same spectral local geometry, the same ecological roles—within that block.

This prediction is what our coherence field framework attempts to test.

1.4 Contributions and Scope

This paper presents:

1. A **theoretical framework** that models genomes as labeled interaction graphs and detects relatedness through spectral-geometric structural correspondence (Section 3).
2. A **comprehensive empirical survey** of 50+ algorithms across three datasets, tracing the evolution from naive k-mer overlap to sophisticated structural methods (Section 2).
3. A **formal analysis** of three theoretical barriers to further progress (Sections 4 and 5).
4. **Three proposed mathematical directions**—optimal transport, persistent homology, and gauge-theoretic field comparison—each representing a qualitative leap beyond current methods (Section 4).
5. An **honest assessment** of what works, what fails, and where the ecosystem approach can or cannot compete with token methods (Section 9).

The computational infrastructure underlying this work is a purpose-built monorepo (`yalumba`) containing 12 TypeScript packages, a custom SPIR-V GPU compilation pipeline, and ∼15,000 lines of code written entirely from scratch with no external bioinformatics libraries.

2 The Empirical Trajectory

Before presenting theory, we describe the empirical program that motivated it. The trajectory from naive k-mer counting to structural ecosystem comparison spans seven months, 50+ algorithm variants, and three datasets of increasing difficulty.

2.1 The Population Sharing Wall

The fundamental challenge for reference-free k-mer methods is that most k-mers are shared between *any* two same-population humans. At $k = 21$, the probability that a randomly chosen 21-mer is identical between two unrelated individuals is:

$$P(\text{match}) = (1 - p_{\text{SNP}})^k \tag{1}$$

where $p_{\text{SNP}} \approx 0.001$ is the per-base SNP rate between two haplotypes from the same population. Substituting:

$$P(\text{match}) = (1 - 0.001)^{21} = 0.999^{21} \approx 0.9792 \tag{2}$$

This means approximately 97.9% of 21-mers are shared between any two individuals from the same population, regardless of kinship. IBD segments contribute an additional $\sim$1–2% of sharing on top of this baseline. The signal-to-noise ratio is 0.01–0.02: a 1–2% excess on a 98% background.

▲ **Limitation**

The population sharing wall. Any method based on counting shared k-mers must distinguish a $\sim$1% signal from a $\sim$98% background. This is feasible with enough data (millions of k-mers per sample) but creates enormous sensitivity to noise, coverage variation, and sequencing error.

2.2 Token-Based Methods: A Complete Survey

We systematically tested every token-based comparison strategy we could conceive. Table 1 summarizes all families with their best results on the CEPH 1463 benchmark (our hardest dataset: three-generation pedigree with four parent–child relationships).

Table 1: Token-based algorithm families tested, with best CEPH 1463 results. Separation = (mean related − mean unrelated) / mean unrelated. Gap = (weakest related − strongest unrelated) / strongest unrelated. A positive gap means *all* related pairs score above *all* unrelated pairs—perfect ordinal discrimination.

Family	Best variant	Sep (%)	Gap (%)
Set overlap	Jaccard, containment, MinHash	< 1	< -50
Run-length	Recombination distance	$+12$	-30
Distribution	Bray–Curtis, Morisita–Horn	$+3$	-25
Information	Entropy rate, NCD [13]	$+1.5$	-40
Multi-scale	Multi-scale run weighting	$+180$	$+50$
Rare run-length	P90 with rarity filtering	**+583**	**+300**

The trajectory tells a clear story. Naive methods (Jaccard, containment) fail because of the population sharing wall: shared k-mers are dominated by the universal 98% background. Run-length methods improve by leveraging the spatial structure of sharing: IBD segments create long *runs* of consecutive shared k-mers, while background sharing creates short, fragmented runs. The P90 statistic (90th percentile of run lengths) captures this efficiently.

The rare-run P90 achieves 583% separation—two orders of magnitude above the baseline.

The breakthrough came from combining run-length analysis with *rarity filtering*: restricting attention to k-mers that appear in fewer than 50% of samples. This automatically removes the population baseline, and the remaining rare k-mers are overwhelmingly informative about kinship.

2.3 The Rarity Insight

The rare-run P90 method works as follows:

1. Identify k-mers shared between a pair of samples.
2. Filter to k-mers that appear in fewer than half the cohort ("rare" k-mers).
3. Compute the lengths of consecutive runs of shared rare k-mers along each read.
4. Take the 90th percentile of run lengths as the similarity score.

▶ **Key Finding**

Rarity filtering is mathematically equivalent to automatic variant detection. A k-mer that appears in fewer than 50% of samples from the same population is, with high probability, tagging a variant (SNP, indel, or structural variant) that is not fixed in the population. The rare-run P90 is therefore detecting runs of shared rare variants—which is precisely what IBD detectors do, but without alignment or variant calling.

This insight reveals both the power and the limitation of the approach. The rare-run P90 achieves +583% separation because it is essentially performing reference-free IBD detection. But it is doing *exactly what GERMLINE does*, just without the alignment step. It does not introduce any fundamentally new signal.

2.4 The Convergence Problem

As token methods improve, they converge toward classical IBD proxies:

- Jaccard → MinHash → weighted MinHash ≈ *Mash* [19]
- Run-length → weighted runs → rare-filtered runs ≈ *reference-free GERMLINE*
- Distribution similarity → IDF-weighted cosine ≈ *TF-IDF* [22] *for genomes*

Each improvement makes the token method more closely approximate what a traditional IBD detector computes. This is *not* a criticism—the methods work precisely because IBD is the correct biological signal. But it means that the token paradigm has a known ceiling: it cannot detect anything that IBD-based methods cannot also detect (given sufficient alignment and phasing quality).

The token paradigm converges toward classical IBD.
The question is whether the ecosystem paradigm can detect something else.

2.5 Structural Methods: A Different Signal

In parallel with the token program, we developed algorithms based on the ecosystem paradigm. These methods construct module-level representations of each genome and compare their organizational structure. Table 2 summarizes results.

Table 2: Structural algorithm families on CEPH 1463. All detect a genuinely different signal from token methods but achieve weaker separation.

Algorithm	Sep (%)	Gap (%)	Status
Module Persistence v3	+3.36	+0.13	**PASSES**
Coalition Transfer v3	+1.97	+0.06	**PASSES**
Module Stability v4	+4.90	−1.04	Fails weakest
Spectral Ecology v3	+4.93	−54.67	Fails
Spectral Ecology v4	+0.02	−4.99	Spouse broken
Spectral Ecology v5	−0.74	−11.29	Scale consist.
Spectral Ecology v6	n/a	−3.21	Persistence
Spectral Ecology v7	n/a	−0.61	Sinkhorn

Module Persistence v3 and Coalition Transfer v3 achieve positive weakest gap—meaning *every* parent–child pair scores above *every* unrelated pair—but with separation orders of magnitude smaller than the rare-run P90 (+3.36% vs. +583%).

▶ **Key Finding**

Structural methods detect a real but weak signal. The positive gap of Module Persistence (+0.13%) and Coalition Transfer (+0.06%) on CEPH 1463 demonstrates that organizational structure carries kinship information independent of token identity. But the signal-to-noise ratio is ∼100× worse than token methods.

2.6 The Spectral Ecology Arc: v1 through v7

The Spectral Ecology family represents our most sustained attempt to extract structural signal through spectral-geometric methods. Seven versions trace a coherent intellectual trajectory, each version's failure directly motivating the next.

Table 3: The Spectral Ecology research trajectory. Each version addresses a specific limitation of its predecessor. Gap = weakest parent–child minus strongest non-parent–child, as percentage.

Ver	Paradigm	Sep	Gap	Sp.	Lesson learned
v1	Sequential adjacency	−0.7%	−26.9%	#1	Graphs too sparse (2–5 edges)
v2	Within-sample spectra	+1.6%	−22.5%	#1	Spectra have ceiling
v3	Cross-sample transport	+4.9%	−54.7%	#1	Transport indiscriminate
v4	Single-scale coherence field	+0.02%	−5.0%	#8	**Spouse confound broken**
v5	Multi-scale fields	−0.7%	−11.3%	#6	Scale consistency real
v6	Persistence fields	n/a	−3.2%	mid	Persistence narrows gap
v7	Sinkhorn + cooperative	n/a	−0.6%	mid	**Near parity**

The critical transition occurs at v4, when the *coherence field* is introduced. Versions v1–v3 attempt to compare ecosystem *summaries* (graph spectra, heat kernel traces, bipartite transport scores). These summaries capture some structural information but cannot distinguish spouses (who have similar global graph structure due to similar coverage) from parent–child pairs (who share specific local neighbourhood arrangements due to IBD).

Seven versions. The weakest gap improved from -26.9% to -0.6%—a $44\times$ reduction.

Version v4 introduces a continuous field $\Phi : G_A \to G_B$ that maps every module in sample A to a position in sample B's ecosystem space, scored by whether the mapping preserves local neighbourhood geometry. This is the first method to break the spouse confound: spouses have globally similar ecosystems but locally dissimilar neighbourhoods, producing incoherent fields with high distortion.

Versions v5–v7 refine the field framework: multi-scale consistency (v5), topological persistence (v6), and Sinkhorn-regularized optimal transport with cooperative stability weighting (v7). The weakest gap contracts from -26.87% (v1) to -0.61% (v7), a $44\times$ improvement that demonstrates convergence toward the passing threshold.

3 Theoretical Framework

We now formalize the ecosystem comparison approach. The framework proceeds in four stages: genome representation, spectral analysis, ecological characterization, and structural correspondence.

3.1 Genome as Labeled Interaction Graph

- **Definition**

Genomic ecosystem graph. Given a set of sequencing reads from an individual, we construct a labeled interaction graph

$$\mathcal{G} = (V, E, w, f)$$

where:

- $V = \{m_1, \ldots, m_n\}$ is a set of *modules*—clusters of co-occurring k-mers identified by community detection on the k-mer co-occurrence graph;
- $E \subseteq V \times V$ is a set of *co-occurrence edges* connecting modules that appear together in sequencing reads more often than expected under independence;
- $w : E \to \mathbb{R}_{>0}$ assigns each edge a weight proportional to the strength of co-

occurrence;

- $f : V \to \mathbb{R}^d$ assigns each module a d-dimensional feature vector encoding its local structural properties (support count, diversity, connectivity, spectral embedding coordinates).

The key departure from token-based representations is that $\mathcal{G}$ encodes *relationships between modules*, not just the modules themselves. Two genomes can share the same set of modules (high token overlap) but arrange them differently (low structural correspondence), or vice versa.

Module construction follows a three-stage pipeline:

1. **Motif extraction.** Extract all canonical k-mers (lexicographically smaller of forward/reverse complement) from reads using a Rabin–Karp rolling hash [11].

2. **Co-occurrence graph.** Build a weighted graph where nodes are frequent k-mers and edges connect k-mers that co-occur within a genomic window (typically 50–150 bp).

3. **Community detection.** Identify densely connected subgraphs as modules using cohesion-based clustering (minimum support and minimum internal edge density thresholds).

In our implementation, typical parameters are $k = 15$, window size 50–150 bp, minimum support 3–5, and minimum cohesion 0.25–0.30. This produces 300–700 modules per sample in the CEPH 1463 dataset, with 200–1,400 co-occurrence edges depending on sequencing depth.

3.2 The Laplacian and Its Spectrum

The organizational structure of $\mathcal{G}$ is analyzed through its graph Laplacian. Let $A \in \mathbb{R}^{n \times n}$ be the weighted adjacency matrix and $D = \mathrm{diag}(d_1, \ldots, d_n)$ the degree matrix where $d_i = \sum_j A_{ij}$.

- **Definition**

Normalized Laplacian. The symmetric normalized Laplacian is

$$L_{\mathrm{sym}} = I - D^{-1/2} A D^{-1/2} \tag{3}$$

with eigendecomposition $L_{\mathrm{sym}} = \Phi \Lambda \Phi^T$, where $\Lambda = \mathrm{diag}(\lambda_0, \lambda_1, \ldots, \lambda_{n-1})$ with $0 = \lambda_0 \leq \lambda_1 \leq \cdots \leq \lambda_{n-1} \leq 2$.

The spectrum $\{\lambda_i\}$ encodes global structural properties:

- $\lambda_0 = 0$ always (constant eigenvector; connected components).
- λ_1 = *algebraic connectivity* (Fiedler value [6]): measures how well-connected the graph is. Small λ_1 indicates a bottleneck or near-disconnection.
- The multiplicity of $\lambda_0 = 0$ equals the number of connected components.
- The spectral gap $\lambda_1 - \lambda_0 = \lambda_1$ controls the mixing time of random walks on the graph.

For ecosystem graphs with 300–700 nodes, we compute the full spectrum via Jacobi eigendecomposition in sub-second time. No iterative approximations are needed.

The *random walk normalized Laplacian* $L_{\text{rw}} = I - D^{-1}A$ is related by $L_{\text{rw}} = D^{1/2} L_{\text{sym}} D^{-1/2}$ and shares the same eigenvalues. We use L_{sym} for eigendecomposition (since it is symmetric and amenable to standard solvers) and interpret results through the random walk perspective.

3.3 Heat Kernel Signature as Ecosystem Fingerprint

The heat kernel on $\mathcal{G}$ describes how a unit of "heat" placed at a node diffuses through the graph over time. It is defined as the matrix exponential of the Laplacian:

$$K_t = \exp(-tL_{\text{sym}}) = \sum_{k=0}^{n-1} e^{-t\lambda_k}\, \phi_k \phi_k^T \tag{4}$$

where ϕ_k is the k-th eigenvector. The *heat kernel signature* (HKS) of node v at timescale t is:

$$\text{HKS}(v,t) = K_t(v,v) = \sum_{k=0}^{n-1} e^{-t\lambda_k}\, \phi_k(v)^2 \tag{5}$$

▶ **Key Finding**

The HKS is a multi-scale structural descriptor. At small t, HKS captures the immediate neighbourhood of v (how many edges, how dense). At large t, HKS captures v's global position in the graph (central versus peripheral). The full signature $\text{HKS}(v,\cdot) : \mathbb{R}_{>0} \to \mathbb{R}$ is a complete characterization of v's structural role up to isospectral ambiguity.

We evaluate the HKS at 10 logarithmically spaced timescales from $t = 0.1$ to $t = 100$, producing a 10-dimensional descriptor per module. The graph-level trace $\sum_v \text{HKS}(v,t) = \text{trace}(K_t) = \sum_k e^{-t\lambda_k}$ is a permutation-invariant graph fingerprint used in our within-sample

spectral comparison (Spectral Ecology v2).

3.4 Ecological Role Assignment via Spectral Embedding

The first d eigenvectors of L_{sym} define a spectral embedding $\psi : V \to \mathbb{R}^d$ where $\psi(v) = (\phi_1(v), \ldots, \phi_d(v))$. This embedding maps structurally similar nodes to nearby points in $\mathbb{R}^d$, regardless of their identity (which k-mers they contain).

We classify modules into ecological roles based on their spectral embedding coordinates and graph-theoretic properties:

Table 4: Ecological role taxonomy for genomic modules. Classification is based on spectral embedding coordinates and local graph properties.

Role	Graph property	Biological interpretation
Core	High degree, central embedding	Conserved genomic structure
Satellite	Low degree, peripheral embedding	Variant-rich region
Bridge	High betweenness, mid embedding	Connects genomic domains
Scaffold	High clustering coefficient	Dense local structure
Boundary	Spectral gap position	Domain transition zone

The role distribution of a genome—e.g., "8 cores, 85 satellites, 356 bridges, 12 scaffolds, 4 boundaries"—is a coarse structural fingerprint that is independent of which k-mers populate each module. Two genomes can have identical role distributions despite sharing no k-mers.

3.5 Coherence Fields: The Key Theoretical Object

The central theoretical contribution of our framework is the *coherence field*: a continuous mapping between ecosystem spaces that measures structural correspondence.

- **Definition**

Coherence field. Given two genomic ecosystem graphs $\mathcal{G}_A$ and $\mathcal{G}_B$ with patch tensor representations $\{t_i^A\}_{i=1}^{n_A} \subset \mathbb{R}^d$ and $\{t_j^B\}_{j=1}^{n_B} \subset \mathbb{R}^d$, the coherence field $\Phi : \mathcal{G}_A \to \mathbb{R}^d$ maps each module in A to a position in B's ecosystem space via kernel regression:

$$\Phi(m_i^A) = \frac{\sum_{j=1}^{n_B} K(t_i^A, t_j^B) \cdot t_j^B}{\sum_{j=1}^{n_B} K(t_i^A, t_j^B)} \tag{6}$$

where $K(x, y) = \exp\bigl(-\|x - y\|^2/2\sigma^2\bigr)$ is a Gaussian kernel with auto-scaled bandwidth σ equal to the median pairwise distance.

The coherence field is not a hard assignment (each A-module maps to one B-module) but a

soft, continuous mapping. Each A-module maps to a *weighted blend* of B-module positions, with weights determined by structural similarity in patch-tensor space.

The patch tensor $t_i \in \mathbb{R}^d$ (with $d = 13$ in our implementation) encodes the local structural environment of each module:

The coherence field is the ecosystem analogue of sequence alignment. Alignment maps bases to bases; the field maps ecological neighbourhoods to ecological neighbourhoods.

- Local Laplacian eigenvalues $\lambda_1, \ldots, \lambda_6$ of the subgraph induced by the module's 1-hop neighbourhood: the "spectral shape" of the local ecosystem.
- Role composition (5 fractions): the proportion of core, satellite, bridge, scaffold, and boundary modules in the neighbourhood.
- Patch density: fraction of possible edges present in the local subgraph.
- Relative patch size: number of nodes in the patch divided by total graph size.

3.6 Cooperative Stability: Measuring Field Quality

The coherence field alone does not produce a similarity score. We need to measure how *well-behaved* the field is—whether it represents a coherent structural correspondence or a random, meaningless mapping.

• **Definition**

Cooperative stability. The cooperative stability of a coherence field $\Phi : \mathcal{G}_A \to \mathbb{R}^d$ is the smoothness of the mapping with respect to the graph structure of A:

$$S(\Phi) = \exp(-\mathrm{CV}(\{\|\Phi(m_i) - \Phi(m_j)\| : (m_i, m_j) \in E_A\})) \tag{7}$$

where CV denotes the coefficient of variation (standard deviation divided by mean) of the set of edge-mapped distances.

High cooperative stability ($S \to 1$) means that neighbouring modules in A map to nearby positions in B's ecosystem space—the field preserves local neighbourhood structure. Low cooperative stability ($S \to 0$) means the mapping is erratic: neighbours in A map to distant, unrelated positions in B.

▶ **Key Finding**

What cooperative stability actually measures. Consider modules m_i and m_j that are neighbours in genome A (they co-occur in reads, their ecological niches overlap). Cooperative stability asks: "do the corresponding regions of genome B also place these modules' analogues in neighbouring positions?"

> This is NOT asking "do A and B share the same modules" (token matching). It is asking "do A and B *organize* their modules in the same way?"

3.7 Why Cooperative Stability Should Detect Relatedness

The theoretical argument for cooperative stability as a relatedness signal proceeds as follows:

1. **IBD segments are inherited intact.** When a child inherits a haplotype block from a parent, the entire arrangement of motifs, their co-occurrence patterns, and their local spectral geometry within that block is inherited intact (modulo sequencing errors and somatic mutations).

2. **Within IBD segments, arrangement is identical.** The coherence field restricted to modules from an IBD segment should be a near-identity mapping: each A-module maps to the "same" module in B. This produces perfect cooperative stability within the segment.

3. **Between non-IBD segments, arrangement is random.** For genomic regions that are *not* IBD, A and B inherited different haplotype blocks from different ancestors. Even if individual motifs happen to match (due to population sharing), their arrangements are independent. The coherence field over these regions has no reason to be smooth.

4. **Global cooperative stability reflects IBD fraction.** Parent–child pairs share $\sim$ 50% of their genome IBD. The coherence field is smooth over the IBD half and erratic over the non-IBD half. The aggregate smoothness should be higher than for unrelated pairs (where IBD fraction is near zero).

▲ **Limitation**

Why cooperative stability currently does not fully work. The argument above assumes that we can distinguish "smooth over IBD segments" from "smooth by coincidence." In practice, three barriers prevent this (detailed in Section 5):

1. Coverage confound: similar sequencing depth creates similar module counts and graph densities, producing spurious smoothness.
2. Symmetry: computing only $\Phi_{A\to B}$ (not also $\Phi_{B\to A}$) allows directional artifacts.
3. Outlier sensitivity: atypical samples with unusual role distributions (e.g., NA12891 with 356 bridge modules) produce distinctive patch tensors that create artificially consistent fields.

4 Three Proposed Theoretical Directions

The coherence field framework (Spectral Ecology v4–v7) has narrowed the gap from -26.87% to -0.61% but appears to be approaching diminishing returns within the kernel-regression paradigm. We propose three mathematically distinct directions that represent qualitative leaps rather than incremental refinements.

4.1 Gromov–Wasserstein Distance Between Ecosystem Metric Spaces

4.1.1 Motivation

The coherence field Φ constructs an explicit point-to-point mapping between modules. This is sensitive to outliers and requires choosing a kernel bandwidth. An alternative is to compare the *metric spaces* directly, without any correspondence.

- **Definition**

Gromov–Wasserstein distance [16]. Given two metric measure spaces (X, d_X, μ_X) and (Y, d_Y, μ_Y), the Gromov–Wasserstein distance is

$$\mathrm{GW}(X,Y) = \inf_{\pi \in \Pi(\mu_X,\mu_Y)} \left(\iint_{X\times Y} \iint_{X\times Y} |d_X(x,x') - d_Y(y,y')|^2 \, d\pi(x,y)\, d\pi(x',y') \right)^{1/2} \tag{8}$$

where $\Pi(\mu_X, \mu_Y)$ is the set of couplings (transport plans) between μ_X and μ_Y.

In words: GW distance finds the optimal soft correspondence between X and Y that minimizes the distortion of *internal distances*. It asks: "can the internal geometry of X be isometrically mapped to Y?"

4.1.2 Application to Genomic Ecosystems

For genomes A and B:

- $X = \{t_1^A, \ldots, t_{n_A}^A\} \subset \mathbb{R}^{13}$: patch tensors of A's modules.
- $Y = \{t_1^B, \ldots, t_{n_B}^B\} \subset \mathbb{R}^{13}$: patch tensors of B's modules.
- $d_X(i,i') = \|t_i^A - t_{i'}^A\|$: Euclidean distance in patch-tensor space.
- μ_X, μ_Y: uniform measures (all modules equally weighted) or support-weighted measures.

▶ **Key Finding**

Why GW might work. GW distance compares the *shape* of the internal distance distribution, not the positions of individual modules. Related individuals share inherited arrangement patterns—modules that are close together in one genome should be close together in the other—producing similar internal distance distributions. Unrelated individuals share random arrangements, producing different distributions.
Critically, GW is **inherently size-invariant**: it does not require the two ecosystems to have the same number of modules. This directly addresses the coverage confound, where different sequencing depths produce different module counts.

4.1.3 Computational Feasibility

The exact GW problem is NP-hard in general, but efficient algorithms exist for the entropic-regularized version. The Sinkhorn-based solver of Peyré et al. [20] computes GW_ε in $O(n^2 \log n)$ time per outer iteration, with $n = \max(n_A, n_B)$. For ecosystem graphs with 400–700 modules, this is feasible in seconds.

4.1.4 Comparison to Coherence Fields

Table 5: Coherence fields vs. Gromov–Wasserstein distance.

Property	Coherence field	GW distance
Correspondence	Soft (kernel regression)	Optimal (transport plan)
Symmetry	Asymmetric ($A \to B$)	Symmetric by construction
Size invariance	No (module count matters)	Yes (metric space)
Outlier sensitivity	High (kernel dominated)	Moderate (transport)
Metric?	No	Yes (a proper distance)
Complexity	$O(n_A \cdot n_B)$	$O(n^2 \log n)$ per iteration

4.2 Persistent Homology of Co-Occurrence Complexes

4.2.1 Motivation

Spectral methods capture the "shape" of the ecosystem graph through eigenvalues, but they are blind to certain topological features. Two graphs can have identical spectra but different topology (co-spectral non-isomorphic graphs). Persistent homology [8] provides a complementary topological characterization that captures features invisible to the spectrum.

4.2.2 Filtration Construction

Starting from the weighted co-occurrence graph, we construct a filtration of simplicial complexes:

1. **Threshold filtration.** At threshold ε, include all edges with weight $\geq \varepsilon$ and all cliques (complete subgraphs) induced by those edges. As ε decreases from $\max(w)$ to 0, more edges and higher-dimensional simplices appear.

2. **Vietoris–Rips filtration.** Embed modules in patch-tensor space and build the Rips complex at increasing radius. This captures metric-space topology rather than graph topology.

- **Definition**

Persistence diagram. For each topological feature (connected component, loop, void, ...), the persistence diagram records a *birth time* b (the threshold at which the feature first appears) and a *death time* d (the threshold at which the feature is destroyed by merger or filling). The persistence $d - b$ measures the significance of the feature: long-lived features represent robust structural properties, while short-lived features are noise.

4.2.3 What Persistent Homology Captures

- H_0 (**connected components**): how the ecosystem fragments into clusters as edge weight threshold increases. Long-lived H_0 features correspond to robust subcommunities of modules.

- H_1 (**loops**): cycles in the co-occurrence graph that persist across thresholds. These may indicate regulatory circuits, repeat structures, or circular dependencies between modules.

- H_2 (**voids**): higher-dimensional cavities in the simplicial complex. Rare in genomic data but potentially informative about mesoscale organization.

▶ **Key Finding**

Why persistent homology might work. Persistence diagrams are *stable* under small perturbations (the Bottleneck Stability Theorem [8]): if two ecosystems differ by small noise, their persistence diagrams are close. They are *inherently size-invariant*: the number of features depends on topology, not on the number of nodes. And they capture *complementary information* to spectral methods: two ecosystems with identical spectra can have different persistence diagrams.

4.2.4 Comparison Between Genomes

Given persistence diagrams D_A and D_B for two genomes, we compute their distance using the p-Wasserstein distance between diagrams:

$$W_p(D_A, D_B) = \left(\inf_{\gamma: D_A \to D_B} \sum_{x \in D_A} \|x - \gamma(x)\|_\infty^p \right)^{1/p} \tag{9}$$

where the infimum is over all bijections γ between the (augmented) diagrams. The augmentation allows unmatched points to be paired with their projection onto the diagonal (birth = death), representing ephemeral features.

4.2.5 Computational Cost

Computing the persistence diagram from a weighted graph on n nodes requires $O(n^3)$ time for the boundary matrix reduction (or $O(n^\omega)$ using fast matrix multiplication). For $n = 700$ modules, this is sub-second. The Wasserstein distance between two diagrams with m points is $O(m^3)$ via the Hungarian algorithm, or $O(m^{1+\varepsilon})$ using recent geometric algorithms [12].

4.3 Gauge-Theoretic Field Comparison

4.3.1 Motivation

The coherence field Φ is scored by its pointwise smoothness (cooperative stability) and its pointwise accuracy (curvature mismatch, distortion). Both measures are sensitive to outliers: a few modules with atypical patch tensors can dominate the aggregate score.

We propose reframing the field comparison in terms of *gauge theory*: instead of comparing Φ pointwise, we compare the *connection*—the way Φ varies along edges.

4.3.2 The Field as a Fiber Bundle Section

- **Definition**

Coherence bundle. Let $\mathcal{G}_A = (V_A, E_A)$ be the base graph. Over each node $v \in V_A$, attach a fiber $F_v \cong \mathbb{R}^d$ representing the possible positions in B's ecosystem space. The coherence field Φ is a section of this bundle: $\Phi(v) \in F_v$ for each v.

The *connection* on this bundle is defined by the transition functions along edges:

$$\nabla_{ij}\Phi = \Phi(m_j) - P_{ij}(\Phi(m_i)) \tag{10}$$

where $P_{ij} : F_i \to F_j$ is the parallel transport operator (in the simplest case, the identity; in a more sophisticated version, the optimal rotation/scaling from the tangent space at i to the tangent space at j).

4.3.3 Curvature as a Relatedness Invariant

The *curvature* of the connection measures the holonomy around closed loops in the graph. For a triangle (i, j, k) in $\mathcal{G}_A$:

$$R_{ijk} = \nabla_{jk} + \nabla_{ki} + \nabla_{ij} = \Phi(k) - P_{jk}(\Phi(j)) + \Phi(i) - P_{ki}(\Phi(k)) + \Phi(j) - P_{ij}(\Phi(i)) \quad (11)$$

If the parallel transport is trivial ($P = \mathrm{id}$), this simplifies to the "boundary" of Φ around the triangle.

▶ **Key Finding**

Why curvature should detect relatedness. Related individuals have coherence fields with *low curvature* (flat connections): the inherited organizational structure induces a *consistent* mapping that "commutes" around loops. If modules $i \to j \to k$ form a triangle in A's ecosystem, and their images $\Phi(i), \Phi(j), \Phi(k)$ form a corresponding triangle in B's ecosystem space, then the holonomy around the loop is zero. Unrelated individuals produce *high-curvature* fields: the mapping is locally reasonable (each edge maps smoothly) but globally inconsistent (going around a loop returns to a different point).

4.3.4 Aggregate Curvature Score

The total curvature is aggregated over all triangles in $\mathcal{G}_A$:

$$\mathcal{R}(\Phi) = \frac{1}{|T(\mathcal{G}_A)|} \sum_{(i,j,k) \in T(\mathcal{G}_A)} \|R_{ijk}\|^2 \quad (12)$$

where $T(\mathcal{G}_A)$ is the set of triangles in A's graph. Low $\mathcal{R}$ indicates a flat (consistent) field; high $\mathcal{R}$ indicates an inconsistent field.

The relatedness score is then $S_{\text{gauge}} = \exp(-\alpha \cdot \mathcal{R}(\Phi))$ for an appropriate scaling constant α.

▲ **Limitation**

This is the most speculative direction. Gauge-theoretic methods have been applied to image analysis (shape matching via connections on fiber bundles [18]) and to network analysis (Ollivier–Ricci curvature on graphs), but not to genomic ecosystem comparison. The key open question is whether genomic ecosystem graphs contain enough triangles to make curvature estimation statistically meaningful. In our CEPH 1463 data, typical ecosystem graphs have 300–700 modules and 200–1,400 edges, yielding $O(10^2)$–$O(10^3)$ triangles—likely sufficient, but empirical validation is needed.

4.3.5 Comparison of the Three Directions

Table 6: Summary of three proposed theoretical directions.

Direction	Size-inv.	Outlier	Metric?	Theory basis
GW distance	Yes	Moderate	Yes	Optimal transport
Persistent homology	Yes	Low	Yes	Algebraic topology
Gauge curvature	Partial	Low	No	Differential geometry

We believe these directions are complementary: GW distance captures global metric structure, persistent homology captures topological invariants, and gauge curvature captures local consistency. A combined score incorporating all three would leverage multiple mathematical perspectives on the same underlying question: "do these two ecosystems have the same organizational structure?"

5 The Coverage Confound: A Formal Analysis

The single largest obstacle to structural relatedness detection is confounding by sequencing coverage. This section provides a formal analysis of why coverage creates spurious similarity and proposes four deconfounding strategies.

5.1 Why Coverage Creates Spurious Similarity

• **Definition**

Null model. Consider two genomes with *identical* k-mer frequency distributions but *independent* sequencing. Let c_A and c_B denote their sequencing coverage (average number of reads covering each base).

Under this null model, the ecosystem graph properties scale predictably with coverage:

1. **Module count.** A k-mer becomes a "module motif" when it exceeds a support threshold τ. The probability that a motif with population frequency p exceeds threshold τ at coverage c is approximately $P(\text{Poisson}(p \cdot c) \geq \tau)$. Thus module count $n \sim F(c)$ where F is a monotone increasing function of coverage.

2. **Edge count.** Two motifs co-occur in a window when both are sequenced. If motifs have independent coverage, the expected co-occurrence count scales as $O(c^2)$, and the number of edges exceeding a co-occurrence threshold scales as $O(c^2)$ as well.

3. **Spectral gap.** For Erdős–Rényi random graphs $G(n, p)$ with edge probability $p \propto c^2/n^2$, the spectral gap $\lambda_1 \to 1$ as $np \to \infty$. Higher coverage produces denser graphs with larger spectral gaps.

4. **Heat kernel trace.** The normalized trace $\text{trace}(K_t)/n = \sum_k e^{-t\lambda_k}/n$ depends on the eigenvalue distribution, which itself depends on coverage through graph density.

▶ **Key Finding**

All standard spectral features correlate with coverage. Under the null model, samples with similar coverage produce similar module counts, edge counts, spectral gaps, and heat kernel traces—regardless of kinship. This explains why spouses (who often have similar sequencing depth due to being processed together) are difficult to separate from parent–child pairs.

5.2 Formal Scaling Relations

Let c denote coverage, $n(c)$ the module count, $m(c)$ the edge count, and $\lambda_1(c)$ the spectral gap. Under our null model:

$$n(c) \approx N_0 \cdot \left(1 - e^{-\alpha c}\right) \tag{13}$$

$$m(c) \approx M_0 \cdot \left(1 - e^{-\beta c^2}\right) \tag{14}$$

$$\lambda_1(c) \approx 1 - \sqrt{\frac{n(c)}{m(c)}} \cdot \lambda_1^{(\mathrm{reg})} \tag{15}$$

where N_0 is the total number of possible modules, M_0 the total number of possible edges, and α, β are detection sensitivity parameters. Equation (13) saturates as coverage increases (all possible modules are eventually detected); equation (14) saturates faster due to the quadratic dependence on coverage.

The implication is that *any ecosystem feature that depends monotonically on module count, edge count, or graph density will be confounded by coverage.* This includes most spectral features, most graph statistics, and most straightforward comparisons.

5.3 Proposed Deconfounding Strategies

We propose four strategies, ordered by increasing mathematical sophistication.

5.3.1 Strategy 1: Degree-Sequence Conditioning

Compare the observed ecosystem to a *null model* with the same degree sequence. Specifically, for each sample, generate B random graphs from the Chung–Lu model with the same expected degree sequence as the observed ecosystem. Compute the spectral features of each random graph. The *deconfounded feature* is the z-score of the observed feature relative to the null distribution:

$$z(f) = \frac{f_{\mathrm{obs}} - \mathbb{E}[f_{\mathrm{null}}]}{\mathrm{Std}[f_{\mathrm{null}}]} \tag{16}$$

This removes the effect of graph density (which is a function of coverage) while preserving the effect of graph *structure* (which is a function of genomic organization).

5.3.2 Strategy 2: Spectral Ratio Features

Instead of using raw eigenvalues $\lambda_1, \lambda_2, \ldots$, use *ratios*: λ_1/λ_2, λ_2/λ_3, etc. If coverage scales all eigenvalues by a common factor (which is approximately true for dense graphs), ratios are invariant:

$$\frac{\lambda_i(c)}{\lambda_j(c)} \approx \frac{\lambda_i}{\lambda_j} \quad \text{(coverage-independent)} \tag{17}$$

This is the spectral analogue of using allele frequency *ratios* instead of raw counts in population genetics.

5.3.3 Strategy 3: Rank-Based Comparison

Convert all module features to ranks before computing ecosystem similarity. Ranks are invariant under any monotone transformation: if coverage multiplies all features by a constant, their ranks are unchanged. Specifically, for each sample, rank the modules by each feature dimension, then compare rank vectors between samples using Spearman correlation.

5.3.4 Strategy 4: Differential Subsampling

For each pair of samples, progressively subsample reads (at 90%, 80%, ..., 10% of original coverage) and compute ecosystem similarity at each level. The *slope* of the similarity-vs-coverage curve is informative: related pairs should have a *shallower slope* (similarity decays slowly because the shared organizational structure is robust) while unrelated pairs should have a *steeper slope* (similarity decays quickly because the background sharing is fragile).

$$\Delta S = \left.\frac{\partial S}{\partial c}\right|_{c=c_{\text{obs}}} \approx \frac{S(0.9c) - S(0.5c)}{0.4c} \tag{18}$$

▶ **Key Finding**

Differential subsampling converts a confound into a signal. Instead of trying to remove the coverage effect, we *measure* the coverage effect and use its magnitude as a feature. Inherited structure is robust to coverage reduction; coincidental similarity is not.

6 Connection to Classical Theory

6.1 IBD as a Special Case of Ecosystem Correspondence

Identity by descent (IBD) is the classical framework for genomic relatedness [5]. Two individuals share an IBD segment if they inherited identical copies of a chromosomal region from a common ancestor. We show that IBD segments, viewed through the ecosystem lens, correspond to subgraphs with *perfect cooperative stability*.

- **Definition**

IBD in the ecosystem framework. Let $S \subseteq V_A$ be the set of modules in genome A whose constituent k-mers fall within an IBD segment shared with genome B, and let $S' \subseteq V_B$ be the corresponding modules in B. There exists a natural bijection $\phi_S : S \to S'$ that maps each module to its IBD-identical counterpart. For this bijection:

1. ϕ_S preserves all co-occurrence edges within the segment: $(m_i, m_j) \in E_A|_S \iff (\phi_S(m_i), \phi_S(m_j)) \in E_B|_{S'}$.
2. ϕ_S preserves node features: $f_A(m_i) = f_B(\phi_S(m_i))$ up to sequencing noise.
3. The cooperative stability of ϕ_S restricted to S is $S(\phi_S|_S) \approx 1$.

The coherence field Φ can be viewed as an *approximate, soft, global* version of this local bijection. Over IBD segments, Φ should approximate ϕ_S. Over non-IBD segments, Φ has no template to follow and produces incoherent mappings.

IBD is a special case of ecosystem correspondence where the correspondence is exact, local, and identity-based. The ecosystem framework generalizes IBD to *approximate, global, structural* correspondence.

6.2 Relationship to Linkage Disequilibrium

Linkage disequilibrium (LD) measures the non-random association of alleles at different loci in a population. Module co-occurrence in our framework is analogous but distinct:

Table 7: Comparison of linkage disequilibrium and module co-occurrence.

Property	LD	Module co-occurrence
Level	Population (allele frequencies)	Individual (read-level)
Reference	Allele at a locus	Module (cluster of k-mers)
Scale	Chromosomal (cM)	Read-level (100–1000 bp)
Requires ref?	Yes (variant positions)	No (de novo modules)

The ecosystem framework replaces population-level LD with individual-level co-occurrence. This has both advantages (no reference genome, no population panel) and disadvantages (noisier, coverage-dependent).

6.3 What the Ecosystem Approach Might Detect That IBD Cannot

The strongest argument for the ecosystem approach is not that it will beat IBD at detecting parent–child relationships (it probably cannot, given the 100× signal gap) but that it might detect phenomena invisible to token-based IBD.

1. **Ancestral organizational patterns.** Organizational structure is more conserved than sequence identity. Two individuals may share organizational patterns inherited from an ancestor many generations back, even though all specific IBD segments have been broken by recombination.

2. **Population-level structural signatures.** Different populations may have characteristic ecosystem structures (role distributions, spectral signatures) that persist across individuals. This could enable *ancestry inference* from organizational structure.

3. **Structural variants.** Large structural variants (inversions, translocations, large deletions) disrupt k-mer identity across the breakpoint but may preserve the organizational structure within the rearranged segment. Liao et al. [14] document extensive structural variation in human genomes; the ecosystem approach might be robust to these variants.

4. **Shared regulatory grammar.** Co-occurrence patterns among k-mers may reflect regulatory relationships (promoter motifs near coding motifs, repeat elements near gene boundaries). This "regulatory grammar" is not captured by haplotype identity but might be captured by ecosystem structure.

5. **Metagenomics.** In metagenomic samples (environmental DNA from microbial communities), there is no single reference genome. The ecosystem approach is inherently reference-free and could potentially be extended to compare community structure across samples.

7 Experimental Status

7.1 The CEPH 1463 Benchmark

Our primary benchmark is the CEPH 1463 Platinum Pedigree, a three-generation family sequenced by Illumina as part of the Platinum Genomes project [21].

Pedigree

```
                GENERATION 1 (Grandparents)
Pat.GF (NA12889)      Pat.GM (NA12890)
       \                 /
        \               /
Mat.GF (NA12891)      Mat.GM (NA12892)
       \                 /
        \               /
              GENERATION 2 (Parents)
        Father (NA12877) ---x--- Mother (NA12878)
                        |
                  GENERATION 3
                Child (NA12882)
```

We use six samples (four grandparents plus father and mother) producing $\binom{6}{2} = 15$ pairwise comparisons. Of these:

- **4 parent–child pairs**: Pat.GF→Father, Pat.GM→Father, Mat.GF→Mother, Mat.GM→Mother
- **1 spouse pair**: Father↔Mother
- **4 in-law pairs**: each grandparent with the opposite parent
- **6 cross-family pairs**: grandparents from different families

The benchmark is deliberately hard: all individuals are from the same population (CEU/European), so the population sharing wall is maximal. Spouses provide a strong negative control: they share no recent ancestry but were often sequenced with similar protocols and depth.

7.2 Current Leaderboard

Table 8 presents the complete leaderboard of all algorithm families tested on CEPH 1463.

Table 8: Complete algorithm leaderboard on CEPH 1463. Separation and gap as defined in Table 1. Status: PASS = positive gap (all parent–child above all non-parent–child); Fail = negative gap.

Family	Best variant	Sep	Gap	Sp.	Status
Token-based methods					
Rare-run	P90	+583%	+300%	low	**PASS**
Multi-scale run	Weighted	+180%	+50%	low	**PASS**
Run-length	Recombin. dist.	+12%	−30%	mid	Fail
Distribution	Bray–Curtis	+3%	−25%	mid	Fail
Information	NCD	+1.5%	−40%	mid	Fail
Set overlap	Jaccard	<1%	−50%	high	Fail
Structural methods					
Module Persist.	v3	+3.4%	+0.13%	low	**PASS**
Coalition Trans.	v3	+2.0%	+0.06%	low	**PASS**
Module Stability	v4	+4.9%	−1.0%	mid	Fail
Spectral Ecol.	v7	n/a	−0.6%	mid	Fail
Spectral Ecol.	v4	+0.02%	−5.0%	#8	Fail
Spectral Ecol.	v3	+4.9%	−54.7%	#1	Fail

7.3 Infrastructure

The computational infrastructure comprises:

- **12 TypeScript packages**: fastq, genome, kmer, alignment, relatedness, vcf, gpu, spirv, runtime, io, compression, math.
- **2 experiment packages**: symbiogenesis (module-level algorithms), ecology (spectral/-graph algorithms).
- **18 ecology source files**: laplacian, heat-kernel, spectral-invariants, role-assignment, module-embedding, node-features, motif-graph, patch-tensor, coherence-field, multiscale-field, persistence-field, pair-transport, compare, ecosystem-signature, null-model, types, index, test suite.
- **4 symbiogenesis algorithms**: module-persistence, coalition-transfer, module-stability, plus barrel exports and runner.
- **LaTeX pipeline**: custom document class with wine palette, branded typography, build system targeting `tectonic`.
- **GPU pipeline** (stub): TypeScript DSL → SPIR-V IR → C codegen → GPU dispatch, with CPU fallbacks for all kernels.

- ∼15,000 **lines** of TypeScript, all from scratch with no external bioinformatics libraries.

7.4 Three Benchmark Datasets

Table 9: Benchmark datasets used in our empirical program.

Dataset	Relationship	Samples	Difficulty
Synthetic family	Parent-child trio	3	Easy
GIAB Ashkenazi [23]	Parent-child trio	3	Medium
CEPH 1463 [21]	3-gen pedigree	6	Hard

The synthetic family is useful for validating implementations but trivial to separate. The GIAB trio provides real sequencing noise. CEPH 1463 is the definitive benchmark: same population, mixed relationships, coverage variation, and the spouse confound.

8 Mathematical Details

This section provides the formal mathematical machinery underlying our framework, intended for readers seeking rigorous foundations.

8.1 The Population Sharing Probability

We derive the expected k-mer sharing between two individuals from the same population. Let π denote the per-base SNP rate between two random haplotypes.

For a k-mer to be *identical* between two haplotypes, all k positions must match. Assuming SNPs are independently distributed (a simplification that ignores LD):

$$P(k\text{-mer match}) = (1 - \pi)^k \tag{19}$$

For diploid organisms, each individual has two haplotypes. A k-mer is "shared" between individuals A and B if it appears on at least one haplotype of each. Let the four haplotype pairs be (h_A^1, h_B^1), (h_A^1, h_B^2), (h_A^2, h_B^1), (h_A^2, h_B^2). By inclusion-exclusion:

$$\begin{aligned} P(\text{shared}) &= 1 - P(\text{not shared}) \\ &= 1 - P(\text{absent from } A \text{ or absent from } B) \\ &\approx 1 - (1 - (1-\pi)^k)^2 \cdot (1 - (1-\pi)^k)^2 \end{aligned} \tag{20}$$

At $\pi = 0.001$, $k = 21$:

$$\begin{aligned} (1-\pi)^k &= 0.9792 \\ (1 - (1-\pi)^k)^2 &= (0.0208)^2 = 0.000433 \\ P(\text{shared}) &\approx 1 - 0.000433 \approx 0.9996 \end{aligned} \tag{21}$$

Virtually all 21-mers are shared. The discriminative information resides in the ∼0.04% of k-mers that are *not* universally shared.

8.2 Expected Run Length Under IBD

Within an IBD segment, consecutive k-mers are shared unless a sequencing error disrupts one. The expected run length of shared k-mers within an IBD segment of length L bases is:

$$\mathbb{E}[\text{run length} \mid \text{IBD}] \approx \frac{1}{\varepsilon_k} = \frac{1}{1 - (1-\varepsilon)^k} \tag{22}$$

where ε is the per-base sequencing error rate. At $\varepsilon = 0.01$ and $k = 21$:

$$\varepsilon_k = 1 - (1 - 0.01)^{21} = 1 - 0.810 = 0.190 \tag{23}$$

$$\mathbb{E}[\text{run}] \approx \frac{1}{0.190} \approx 5.3 \tag{24}$$

Outside IBD, the expected run length is much shorter, determined by the chance overlap probability at the rare-k-mer level. The run-length distribution is the key discriminative feature exploited by the rare-run P90 method.

8.3 Module Count as a Function of Coverage

Let M be the total number of distinct modules in the genome (a property of the genome, not the sequencing). At coverage c, a module with population frequency p and s constituent k-mers is *detected* if at least τ of its k-mers are observed. The number of observed k-mers follows $\text{Binomial}(s, 1-(1-p)^c) \approx \text{Binomial}(s, 1-e^{-pc})$ for large c.

The expected number of detected modules is:

$$n(c) = \sum_{m=1}^{M} P\big(\text{Bin}(s_m, 1-e^{-p_m c}) \geq \tau\big) \tag{25}$$

For typical parameters ($p_m \approx 0.5$, $s_m \approx 10$, $\tau = 3$), this saturates at $c \approx 10\times$, well within standard whole-genome sequencing depth.

8.4 Gromov–Wasserstein: Entropic Regularization

The exact GW problem (Equation 8) is computationally intractable for large n. The entropic-regularized version adds a penalty $\varepsilon\, H(\pi)$ where $H(\pi) = -\sum_{ij} \pi_{ij} \log \pi_{ij}$ is the entropy of the transport plan:

$$\text{GW}_\varepsilon(X, Y) = \inf_{\pi \in \Pi(\mu_X, \mu_Y)} \left[\sum_{i,i'} \sum_{j,j'} |d_X(x_i, x_{i'}) - d_Y(y_j, y_{j'})|^2\, \pi_{ij} \pi_{i'j'} - \varepsilon\, H(\pi) \right] \tag{26}$$

This can be solved by alternating between:

1. Fixing π and solving the linear optimal transport problem (Sinkhorn iterations [7]).
2. Fixing the cost matrix and updating π via a conditional gradient step.

Each Sinkhorn step is $O(n^2)$; typically 10–50 outer iterations suffice for convergence. Total cost: $O(n^2 \cdot K_{\text{Sinkhorn}} \cdot K_{\text{outer}})$ where $K_{\text{Sinkhorn}} \approx 50$ and $K_{\text{outer}} \approx 20$. For $n = 700$: approximately $700^2 \times 50 \times 20 = 4.9 \times 10^8$ operations, feasible in seconds on a modern CPU.

8.5 Persistence Diagram Stability

The Bottleneck Stability Theorem guarantees that small perturbations of the input produce small changes in the persistence diagram.

- **Definition**

Bottleneck Stability Theorem (Edelsbrunner et al. [8]). Let f and g be two tame functions on a topological space X, and let $\mathrm{Dgm}(f)$ and $\mathrm{Dgm}(g)$ denote their persistence diagrams. Then:

$$d_B(\mathrm{Dgm}(f), \mathrm{Dgm}(g)) \leq \|f - g\|_\infty \tag{27}$$

where d_B is the bottleneck distance between diagrams.

For genomic ecosystems, this means: if two samples have similar co-occurrence weight functions (because they share inherited organizational structure), their persistence diagrams will be close in bottleneck distance. Sequencing noise, which adds small perturbations to edge weights, produces correspondingly small changes in the diagram.

8.6 Connection Curvature on Graphs

We formalize the gauge-theoretic direction. Let $\mathcal{G} = (V, E)$ be a graph with a vector field $\Phi : V \to \mathbb{R}^d$ (the coherence field). Define the *discrete connection* ω along an edge $(i, j) \in E$ as:

$$\omega_{ij} = \Phi(j) - \Phi(i) \tag{28}$$

The *discrete curvature* around a triangle (i, j, k) is the holonomy:

$$\Omega_{ijk} = \omega_{ij} + \omega_{jk} + \omega_{ki} = [\Phi(j) - \Phi(i)] + [\Phi(k) - \Phi(j)] + [\Phi(i) - \Phi(k)] = 0 \tag{29}$$

This is identically zero! The trivial connection has zero curvature. To obtain a non-trivial curvature, we must introduce a non-trivial parallel transport. The natural choice is:

$$\omega_{ij}^{\mathrm{OT}} = \Phi(j) - T_{ij}(\Phi(i)) \tag{30}$$

where T_{ij} is the *optimal transport map* from the local neighbourhood of i (in B-space) to the local neighbourhood of j (in B-space). The curvature then measures whether going around a loop in A's graph and applying the local transport maps at each step returns to the starting point in B's space.

- **Definition**

Holonomy curvature. For a triangle $(i, j, k) \in T(\mathcal{G}_A)$:

$$\Omega_{ijk}^{\mathrm{OT}} = \omega_{ij}^{\mathrm{OT}} + \omega_{jk}^{\mathrm{OT}} + \omega_{ki}^{\mathrm{OT}} \tag{31}$$

A "flat" field ($\Omega \approx 0$) indicates that the local transport maps are globally consistent—the structural correspondence between A and B is coherent. A "curved" field ($\|\Omega\|$ large) indicates global inconsistency.

This is directly analogous to the curvature of a connection on a principal bundle in differential geometry [18], discretized to the graph setting.

9 Open Questions

We close by identifying the most important open questions, both theoretical and empirical, that this research program must address.

9.1 Is the Structural Signal Strong Enough?

The central empirical question: is the organizational structure of a genome sufficiently different between related and unrelated individuals to be practically useful? Our current results suggest the signal exists (+3.36% separation for Module Persistence v3, positive weakest gap) but is 100× weaker than the token-level signal (+583% for rare-run P90).

There are two possible interpretations:

1. **The signal is weak because our methods are immature.** Better mathematical tools (GW distance, persistence, gauge curvature) might extract orders of magnitude more structural signal.

2. **The signal is inherently weak because organizational structure varies less than sequence identity.** Related and unrelated individuals may simply have more similar organizational structures than they have similar sequence content.

Distinguishing these hypotheses is critical. If (2) is correct, the ecosystem approach has fundamental limits for close-relationship detection. It might still be valuable for distant-relationship detection, ancestry inference, or metagenomics, where token-level signal is insufficient.

9.2 Can Spectral Methods Match Rare-Run Performance?

Specifically for parent–child detection on CEPH 1463: can any spectral or structural method achieve +50% separation (currently the second-best token result)? Or is the spectral approach better suited for different tasks?

We conjecture that spectral methods will never match rare-run P90 at parent–child detection, because rare-run is essentially doing IBD detection and IBD is the *strongest possible signal* for close relationships. The spectral approach should instead target:

- Distant relationships (beyond the IBD detection horizon)
- Population structure (where organizational patterns may be shared across many individuals)
- Metagenomic community structure (where no reference exists)

9.3 Does the Ecosystem Representation Contain Novel Information?

A critical theoretical question: does the ecosystem graph $\mathcal{G}$ encode information that is *not* present in the k-mer frequency spectrum $\{(k_i, c_i)\}$ (the multiset of k-mers and their counts)?

We argue yes: the ecosystem graph encodes *co-occurrence structure* (which k-mers appear together in reads) which is not captured by the marginal frequency spectrum. Two genomes can have identical k-mer frequency spectra but different co-occurrence graphs if the k-mers are arranged differently along the chromosomes.

However, this argument depends on read length. With infinitely long reads, the co-occurrence graph would capture the entire chromosomal arrangement and contain maximal information. With short reads ($\sim$150 bp for Illumina), the co-occurrence graph captures only *local* arrangement within a 150 bp window. The information content is therefore bounded by the ratio of read length to genome size.

9.4 Information-Theoretic Limits

What is the information-theoretic limit of reference-free relatedness detection from raw reads? Formally: given N reads of length L from a genome of size G at coverage $c = NL/G$, what is the mutual information between the read set and the relatedness status (related/unrelated)?

This is an open theoretical question. A lower bound is provided by the token-level information (the rare-run P90 signal). An upper bound is provided by the total information in the read set about the donor genome. The gap between these bounds determines how much room exists for structural methods to improve upon token methods.

9.5 Extension to Metagenomics

The ecosystem framework is inherently reference-free and operates on communities of interacting motifs. This makes it conceptually applicable to metagenomics, where the "genome" is a community of organisms and the "modules" are shared gene families or metabolic pathways.

The open question is whether the spectral-geometric tools developed for single-organism relatedness detection transfer to community-level comparison. The scale is different (millions of species vs. hundreds of modules), but the mathematical framework—interaction graphs, Laplacian spectra, coherence fields—is the same.

10 Conclusion

10.1 An Honest Assessment

The token paradigm works. On the hardest benchmark we have (CEPH 1463, same-population, three-generation pedigree), the rare-run P90 achieves +583% separation with every parent–child pair scoring decisively above every non-parent–child pair. This is a practical, deployable result that validates reference-free relatedness detection as an engineering possibility.

The ecosystem paradigm is promising but unproven. Module Persistence v3 and Coalition Transfer v3 achieve positive weakest gap on CEPH 1463, demonstrating that organizational structure carries kinship information. But the signal is 100× weaker than the token signal, and the spectral ecology program—despite seven iterations of increasing mathematical sophistication—has not yet achieved positive weakest gap.

10.2 What the Trajectory Demonstrates

The spectral ecology arc (v1 through v7) demonstrates several important facts:

1. **Structural signal exists.** The improvements from −26.87% to −0.61% are not random; each version targets and addresses a specific failure mode.

2. **The spouse confound is breakable.** v4's breakthrough (spouses from rank 1 to rank 8) proves that structural methods can distinguish "similar because co-sequenced" from "similar because related"—something naive token methods cannot do.

3. **Diminishing returns within the kernel-regression paradigm.** The 44× improvement from v1 to v7 is large, but the rate of improvement is decelerating. Qualitative

mathematical leaps (GW, persistence, gauge theory) are needed.

4. **The computational objects are converging.** The progression tokens → summaries → matchings → fields → multi-scale fields → persistence fields suggests a natural hierarchy. Each level adds more structural information to the comparison.

10.3 The Path Forward

We believe the strongest case for the ecosystem approach will come not from beating rare-run P90 at parent–child detection, but from one of three scenarios:

1. **Detecting distant relationships** beyond the IBD detection horizon, where organizational patterns persist after specific sequence identity has been lost to recombination.

2. **Detecting population structure** from organizational signatures, enabling reference-free ancestry inference without variant calling.

3. **Detecting structural relationships in metagenomics**, where no reference genome exists and the ecosystem framework is the only available paradigm.

The three proposed theoretical directions—Gromov–Wasserstein distance, persistent homology, and gauge-theoretic field comparison—represent our best mathematical candidates for qualitative advances. Each addresses a specific limitation of the current coherence field framework (size dependence, spectral blindness, and outlier sensitivity, respectively), and each draws on mature mathematical theory with strong stability guarantees.

The question is not whether genomes can be compared structurally—they can. The question is whether structural comparison reveals anything that sequence comparison does not.

This paper is a request for feedback. The theoretical framework, the empirical trajectory, and the proposed directions are presented in full so that computational biologists, applied mathematicians, and genomicists can evaluate the approach, identify flaws, and suggest alternatives. We welcome criticism. The strongest possible outcome for this program is a rigorous answer—whether positive or negative—to the question of whether inherited ecosystem structure is a practically useful signal for genomic relatedness detection.

A Algorithmic Catalogue

This appendix provides brief descriptions of all major algorithm variants tested during the empirical program, organized by family.

A.1 Set Overlap Family

Table 10: Set overlap algorithms.

Algorithm	Description	CEPH Sep
k-mer Jaccard	$\lvert A \cap B\rvert / \lvert A \cup B\rvert$	$< 1\%$
Containment	$\lvert A \cap B\rvert / \min(\lvert A\rvert, \lvert B\rvert)$	$< 1\%$
MinHash	Approximate Jaccard via sketching	$< 1\%$
Weighted Jaccard	IDF-weighted set intersection	$\sim$2%

All set overlap methods fail because the population sharing wall ($\sim$98% baseline) dominates the IBD signal ($\sim$1% excess).

A.2 Run-Length Family

Table 11: Run-length algorithms.

Algorithm	Description	CEPH Sep
Recombination distance	Mean run length of shared k-mers	+12%
P90 run length	90th percentile of run lengths	+25%
Weighted runs	Length-weighted run score	+30%
Multi-scale runs	Runs at multiple k values	+180%
Rare P90	P90 on rare (<50% freq) k-mers	**+583%**

The rare-run P90 breakthrough demonstrates that rarity filtering removes the population baseline, leaving only the IBD-informative k-mers.

A.3 Distribution Family

Table 12: Distribution comparison algorithms.

Algorithm	Description	CEPH Sep
Cosine similarity	Frequency vector cosine	+2%
Jensen–Shannon	$\sqrt{\mathrm{JSD}(p_A, p_B)}$	+1.5%
Bray–Curtis [1]	$1 - \sum \lvert a_i - b_i \rvert / \sum (a_i + b_i)$	+3%
Morisita–Horn [17]	Overlap index for count data	+2.5%

Distribution methods provide moderate signal but cannot distinguish local sharing (IBD) from global sharing (population baseline).

A.4 Information-Theoretic Family

Table 13: Information-theoretic algorithms.

Algorithm	Description	CEPH Sep
Entropy rate	Conditional entropy of shared k-mers	+1%
NCD [13]	Normalized compression distance	+1.5%
Mutual information	MI between k-mer count vectors	+0.5%

Information-theoretic methods perform poorly, likely because the compressibility of genomic sequences is dominated by repeat content rather than by kinship-related sharing patterns.

A.5 Structural: Module Persistence Family

Module Persistence constructs a shared module vocabulary across all samples and compares samples by which modules they contain.

- **v1**: Per-sample modules with Jaccard comparison. Fails because module IDs are sample-specific.
- **v2**: Shared vocabulary with edge Jaccard. Edge Jaccard has range 86–89% for all pairs (not discriminative).
- **v3**: Rarity-weighted informative module overlap plus support cosine. +3.36% **separation,** +0.13% **gap. PASSES.**

A.6 Structural: Coalition Transfer Family

Coalition Transfer measures sharing of *module coalitions*: specific combinations of modules that co-occur in individual reads.

- **v1**: Raw coalition Jaccard. Moderate signal but confounded by coverage.
- **v2**: IDF-weighted coalition cosine. Boosted in-laws unexpectedly.
- **v3**: Three-way blend (rarity + exclusive + unweighted cosine). +1.97% **separation,** +0.06% **gap. PASSES.**

A.7 Structural: Module Stability Family

Module Stability uses bootstrap resampling to identify which motifs are *stably organized* into modules across subsamples.

- **v1–v3**: Binary stability filtering with various thresholds. Moderate signal.
- **v4**: Continuous stability-weighted rarity Jaccard. +4.90% **average separation** but −1.04% **weakest gap** (fails weakest parent–child pair).

A.8 Structural: Spectral Ecology Family

The Spectral Ecology family is detailed in Section 2.6 and represents our deepest investigation of spectral-geometric methods. Seven versions span the progression from disconnected sequential-adjacency graphs (v1) through Sinkhorn-regularized optimal transport with cooperative stability weighting (v7).

B CEPH 1463 Detailed Pairwise Analysis

For completeness, we present the complete pairwise score tables for the three best-performing algorithms on CEPH 1463.

B.1 Module Persistence v3

Table 14: Module Persistence v3: complete pairwise scores on CEPH 1463. ▶ = parent–child.

Rk	Pair	Score	Rarity	Cosine
1	▶ Pat.GM → Father	0.3248	34.1%	0.320
2	▶ Mat.GF → Mother	0.3221	33.8%	0.318
3	▶ Pat.GF → Father	0.3195	33.4%	0.316
4	▶ Mat.GM → Mother	0.3148	32.9%	0.312
5	Father ↔ Mother	0.3144	32.8%	0.311
6	Mat.GF ↔ Father	0.3132	32.7%	0.310
7	Pat.GF ↔ Mother	0.3118	32.5%	0.309
8	Pat.GM ↔ Mat.GM	0.3105	32.3%	0.308
9	Pat.GF ↔ Mat.GF	0.3098	32.2%	0.307
10	Mat.GM ↔ Father	0.3091	32.1%	0.307

The margins are razor-thin: the gap between rank 4 (weakest parent–child: 0.3148) and rank 5 (strongest non-parent–child: 0.3144) is only +0.13%. This demonstrates that the structural signal is real but fragile.

All four parent–child pairs rank in the top four. Gap: +0.13%. Razor-thin but correct.

B.2 Coalition Transfer v3

Table 15: Coalition Transfer v3: complete pairwise scores on CEPH 1463.

Rk	Pair	Score	Excl.	FreqCos
1	▶ Pat.GF → Father	0.1524	1.82%	0.394
2	▶ Pat.GM → Father	0.1498	1.75%	0.391
3	▶ Mat.GF → Mother	0.1472	1.68%	0.387
4	▶ Mat.GM → Mother	0.1463	1.65%	0.385
5	Father ↔ Mother	0.1462	1.64%	0.384
6	Mat.GF ↔ Father	0.1455	1.62%	0.383
7	Pat.GF ↔ Mother	0.1448	1.59%	0.381
8	Pat.GM ↔ Mat.GM	0.1441	1.57%	0.380
9	Pat.GF ↔ Mat.GF	0.1438	1.56%	0.379
10	Mat.GM ↔ Father	0.1432	1.54%	0.378

Coalition Transfer achieves the same perfect ordinal ranking as Module Persistence but with even thinner margins (+0.06% gap).

C Notation and Symbols

Table 16: Notation used throughout this paper.

Symbol	Meaning
$\mathcal{G} = (V, E, w, f)$	Genomic ecosystem graph
L_{sym}	Symmetric normalized Laplacian
λ_k	k-th eigenvalue of L_{sym}
ϕ_k	k-th eigenvector of L_{sym}
$K_t = \exp(-tL)$	Heat kernel at timescale t
$\text{HKS}(v, t)$	Heat kernel signature of node v
$\Phi : \mathcal{G}_A \to \mathbb{R}^d$	Coherence field
$S(\Phi)$	Cooperative stability of field Φ
$\text{GW}(X, Y)$	Gromov–Wasserstein distance
$\pi \in \Pi(\mu, \nu)$	Optimal transport plan
$H(\pi)$	Entropy of transport plan
D_A, D_B	Persistence diagrams
$W_p(D_A, D_B)$	p-Wasserstein distance between diagrams
ω_{ij}	Discrete connection along edge (i, j)
Ω_{ijk}	Holonomy curvature around triangle
$\mathcal{R}(\Phi)$	Aggregate curvature of field
c	Sequencing coverage
k	k-mer length
π	Per-base SNP rate
n	Number of modules
m	Number of edges
d	Patch tensor dimension (13)
σ	Kernel bandwidth

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